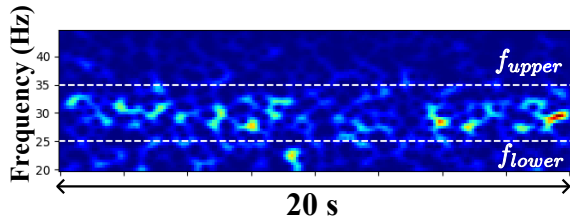
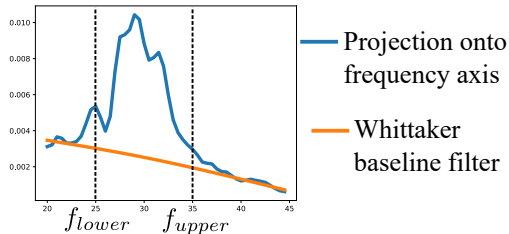


Segmentation of ketamine signature on spectrogram

a) Spectrogram



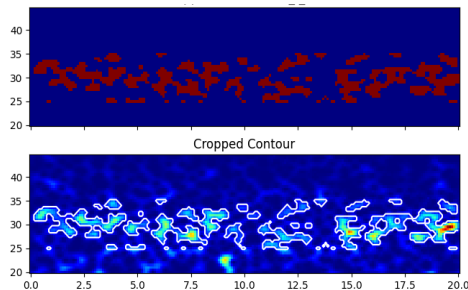
b) Spectrogram projection on frequency



c) Threshold high intensity pixels

$$T(f) = 3 \cdot baseline(f)$$

$$mask(f, t) = \mathbb{I}(\text{spectrogram}(f, t) > T(f))$$
$$f \in [f_{lower}, f_{upper}]$$



d) Extend till background it met

$$T_{background}(f) = 2 \cdot baseline(f)$$

$$mask_{background}(f, t) =$$
$$\mathbb{I}(\text{spectrogram}(f, t) > T_{background}(f))$$

